VENKATESH MALLARAPU

Data Scientist | ML Engineer | MLOps

venkateshmallarapu2@gmail.com | +91 7330648839 | Khammam, India
linkedin.com/in/venkatesh-mallarapu-b69853380 | github.com/venkateshmallarapu2-dot | venkateshmallarapu.vercel.app

PROFESSIONAL SUMMARY

Results-driven Data Scientist and ML Engineer with hands-on expertise in Python, SQL, Machine Learning, and MLOps. Skilled in building end-to-end ML pipelines, CI/CD automation for the ML lifecycle, model governance and lineage, secure REST API model serving, and cloud-based deployment. Projects span financial fraud detection, AI agent systems, and computer vision. Final-year B.Tech Computer Science student with a strong foundation in statistics, algorithms, and database management.

TECHNICAL SKILLS

Programming: Python, SQL
Machine Learning: Regression, Classification, Clustering, Random Forest, SVM, KNN, Naive Bayes, LSTM, XGBoost, Decision Trees
MLOps & CI/CD: CML Workspaces, CI/CD Pipelines for ML Lifecycle (GitHub Actions), Model Versioning, Secure REST API Model Serving
Governance & Monitoring: Model Governance, Lineage Tracking, Drift Detection, Model Monitoring Tools Integration
Libraries & Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow/Keras, Flask, Matplotlib, Seaborn
Tools & Platforms: Jupyter Notebook, Google Colab, Git/GitHub, VS Code
Data Analysis: EDA, Feature Engineering, Data Cleaning, Preprocessing, SMOTE, Outlier Handling
DS & Algorithms: Arrays, Linked Lists, Stacks, Queues, Trees, Graphs, Hash Maps, Dynamic Programming, BFS/DFS
Gen AI & LLMs: Prompt Engineering, RAG Pipelines, LLM Integration (OpenAI/HuggingFace), AI Agents, Code Generation & Translation
Enterprise Search: Elasticsearch (indexing, querying, relevance tuning), Splunk (log analysis, dashboards, alerts)

EXPERIENCE

Data Scientist Intern · Evoastra Ventures *Feb 2026 – Apr 2026*

- Performed EDA, data preprocessing, and feature engineering on real-world business datasets to identify key patterns and inform model design.
- Built and evaluated ML models (Random Forest, SVM, Regression) for AI-based supply chain optimization and demand forecasting, improving prediction accuracy by ~15%.
- Deployed models via REST APIs for real-time inference; implemented CI/CD pipeline steps for ML lifecycle automation enabling reproducible, versioned deployments.
- Integrated model monitoring and governance tools to track performance post-deployment; maintained model lineage documentation across experiment iterations.
- Created dashboards and data visualizations using Matplotlib and Seaborn to communicate actionable insights to stakeholders.

PROJECTS

Credit Card Fraud Detection | *XGBoost, Random Forest, SMOTE, Scikit-learn, Flask, CI/CD Pipeline*

- Developed a real-time financial fraud classification system on highly imbalanced data (fraud < 0.2% of records) using StandardScaler normalization on PCA-transformed features and SMOTE oversampling to balance the training distribution.
- Benchmarked Logistic Regression, XGBoost, and Isolation Forest against a 200-estimator Random Forest Classifier, achieving strong Precision, Recall, and F1-Score (0.95+ PR-AUC) on the minority fraud class.
- Built a CI/CD pipeline for automated model retraining with governance and lineage tracking; served the Pickle-serialized model via a Flask REST API returning fraud/legit labels with confidence scores, rendered on a live JavaScript fraud-monitoring dashboard.

GitHub: github.com/venkateshmallarapu2-dot/credit-card-fraud-detection-using-ml

AI Company Intelligence Agent | *Python, AI Agent (ReAct), LLM, Node.js/Express, REST API, SSE*

- Architected an autonomous AI agent pipeline for deep-dive company research, orchestrating web search, data scraping, financial summary extraction, and report generation via structured LLM prompt chaining.
- Implemented a Reason-Act (ReAct) loop where the agent reasons about missing information, selects the appropriate tool (search, fetch, parse), executes it, and feeds results back into context for the next step.
- Built a Node.js/Express middleware layer to handle session state, tool-invocation routing, and streaming of LLM responses to the frontend via Server-Sent Events; rendered real-time company intelligence cards covering financials, tech stack, hiring trends, and competitive positioning.

GitHub: github.com/venkateshmallarapu2-dot/AI-Company-Intelligence-Agent

Handwritten Character Recognition | *CNN, TensorFlow/Keras, OpenCV, Flask, REST API*

- Designed and trained a deep CNN on the EMNIST Balanced dataset (814,255 images, 62 character classes) using Conv2D, MaxPooling, Batch Normalization, and Dropout layers, achieving ~92% top-1 accuracy with sub-100ms inference latency.
- Applied Keras ImageDataGenerator augmentation (rotation, zoom, shifts) for generalization; used OpenCV for canvas preprocessing (grayscale, Gaussian blur, adaptive thresholding, contour-based bounding boxes).
- Deployed a Flask REST API that receives base64-encoded canvas frames, runs real-time inference, and returns the predicted character with confidence score to an HTML5 Canvas frontend.

GitHub: github.com/venkateshmallarapu2-dot/hand-written-characteristics-recognition-

EDUCATION & CERTIFICATIONS

B.Tech Computer Science · Holy Mary Institute of Technology & Science, Hyderabad *May 2026*

- Machine Learning, Python, Artificial Intelligence — Visualpath Data Science | Data Science — Tutedude